

Save my exam Note (Edexcel level)

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1. Physical Chemistry

1.1. Atomic structure

1.1.1. Structure of the atom

Subatomic particle	Relative charge	Relative mass
Proton	+1	1
Neutron	0	1
Electron	-1	1/1836



A = mass number / nucleon number

Z = atomic number / proton number

Number of neutrons = mass number – atomic number

For neutral atom, the number of electrons will be same as number of protons

1.1.2. Isotopes

Atoms of the same number of proton but different number of neutron.

Example of isotopes:

Carbon-12 and carbon-14

Hydrogen-1 (protium), hydrogen-2 (deuterium), hydrogen-3 (tritium)

Isotopes have same chemical property

They have the same number of electrons in outer shells

Electrons take part in chemical reaction and therefore determine chemistry of atom

but different physical property

because have difference in number of neutrons

causing having difference in mass density, melting point and boiling point

All atoms and ions of the same element will have the same number of protons in the nucleus

Ions are formed when atoms either gain electrons (form positive charged ion) or lose electrons (form negative charged ion)

1.1.3. Relative mass

Relative mass of an atom uses the carbon-12 isotope as international standard

The mass of carbon-12 = 12 amu (atomic mass unit)

Standard mass for 1u = 1/12 mass of carbon-12

Relative atomic mass

Average mass of atom relative to 1/12 mass of carbon-12 atom

Because most elements in the periodic table represent mixture of different isotopes

Relative formula mass is the total mass of the substance

Eg. ionic compound (NaCl)

If the substance is molecule, can use as relative molecular mass

To calculate it, just add up the relative atomic mass of all atoms present in the formula

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